DH Wrap Up

EGR 445/545

GVSU

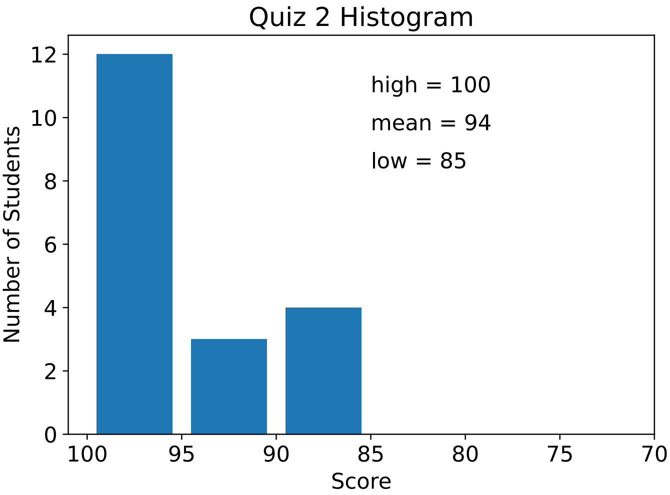
Plan for Today

- quiz 2 solution
- white vs. yellow Fanuc DH discussion
- PUMA DH discussion
- project 1 demos

Reminder

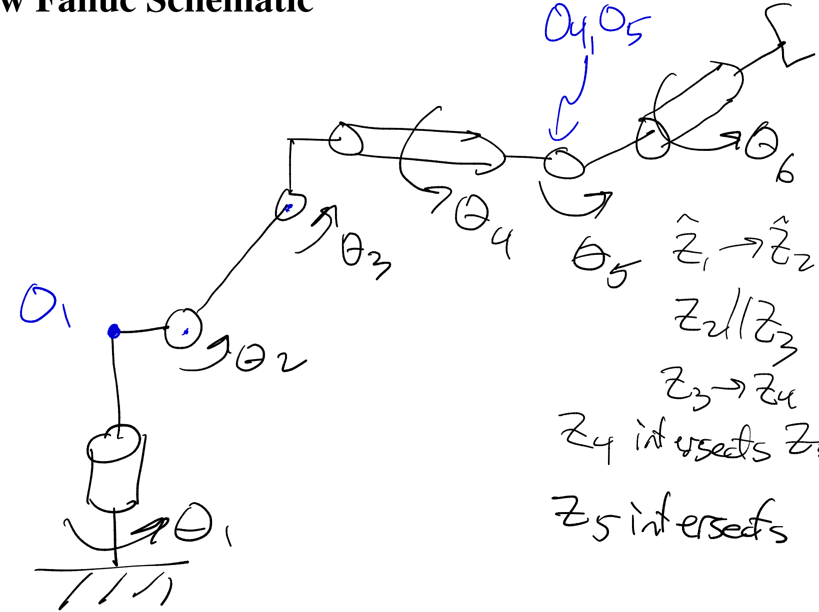
- Exam 1 Monday, 6/14
    - approx. 60 minutes
    - chapters 2 and 3
    - 1-2 conceptual questions
    - 3-4 questions with Python available
        - must start with "official blank" notebook
        - you may use the Python 'help' function
        - you may not use any other resources
    - 1 page of hand written notes
        - 8.5" by 11" US letter paper
            - Front and back

Quiz 2 Histogram



White vs. Yellow Fanuc Discussion

Yellow Fanuc Schematic



$\hat{z}_1 \to \hat{z}_2$ non/non case 3

$z_2 \parallel z_3$ case

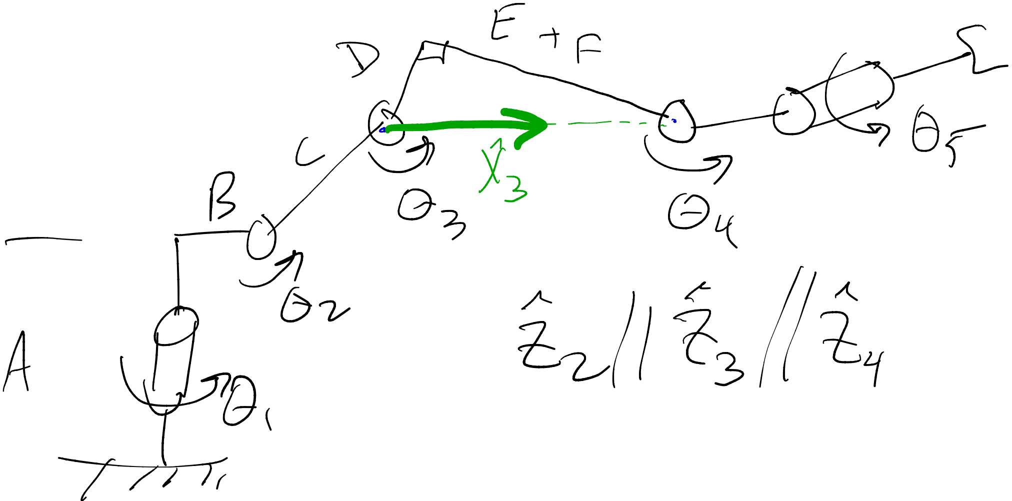
$z_3 \to z_4$ non/non case 3

z_4 intersects z_5 (case 2)

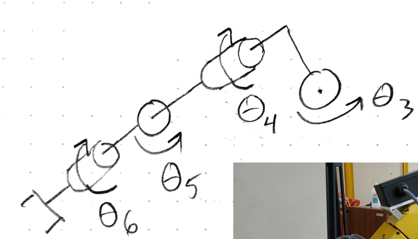
z_5 intersects z_6 case 2

White Fanuc Schematic

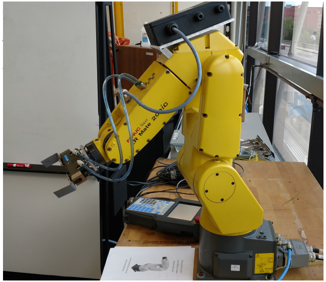
- what is different?



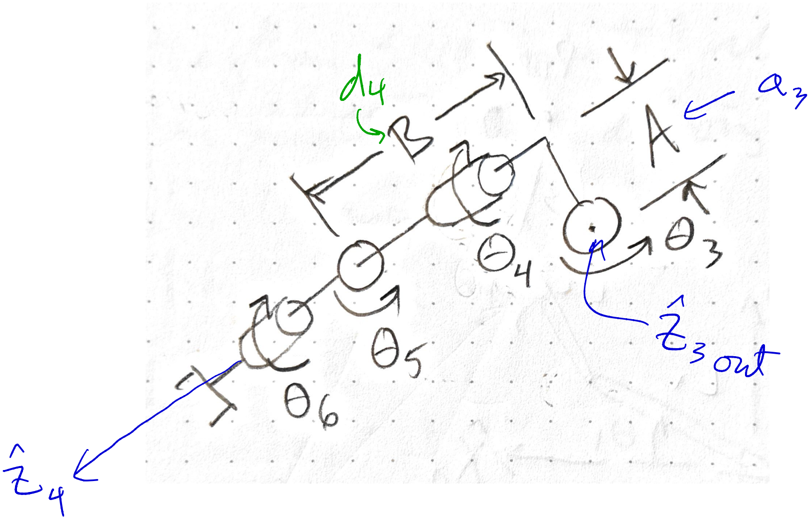
Yellow Fanuc Forearm/Wrist



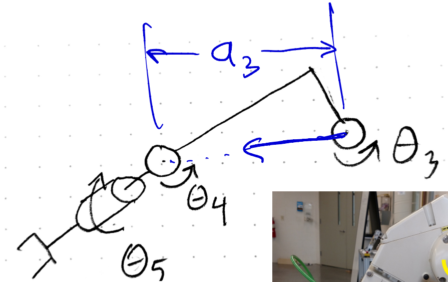
- where is O_3?
- do we have any flexibility in the location of O_3?



Yellow Fanuc: What do we call these two parameters?

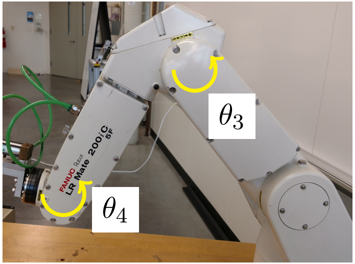


White Fanuc Forearm/Wrist

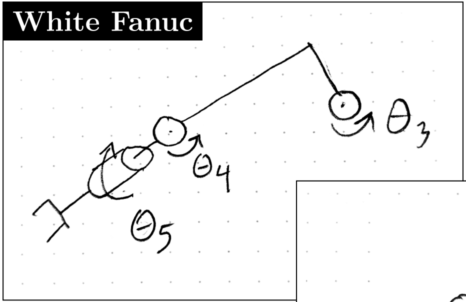


- only 5 joints
- what is different about $\hat{Z}_3$ and $\hat{Z}_4$?
- where is O_3?
- what are a_3 and d_4?

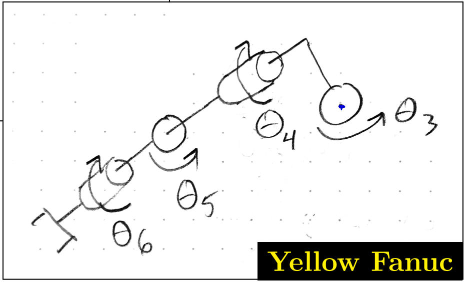
0



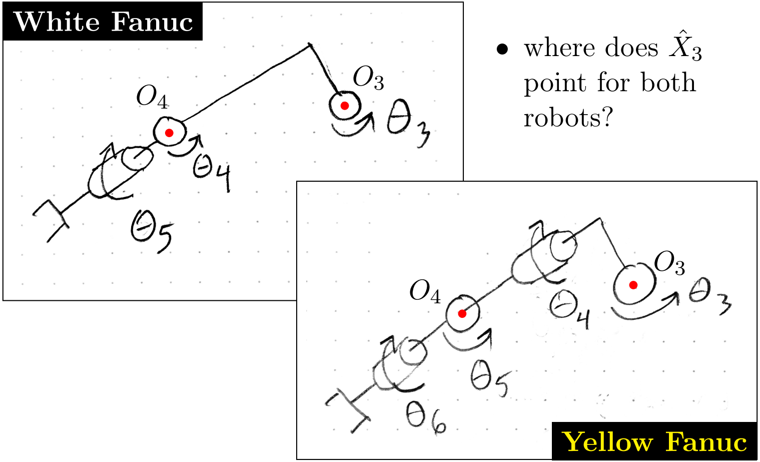
Comparing the White and Yellow Fanucs



- where is O_3 for both robots?
- do we have any flexibility?

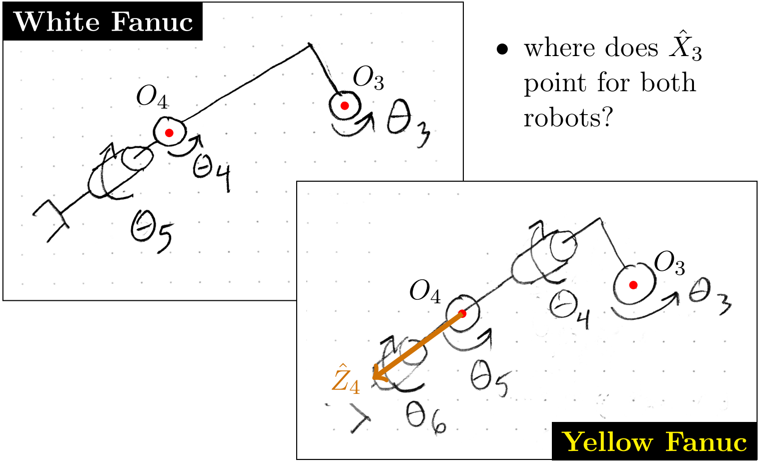


Comparing the White and Yellow Fanucs



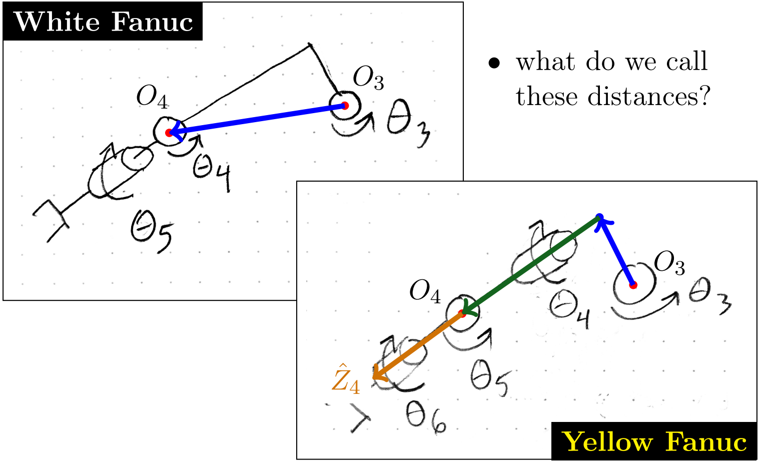
- where does $\hat{X}_3$ point for both robots?

Comparing the White and Yellow Fanucs



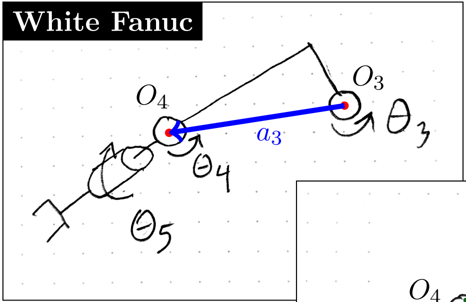
- where does $\hat{X}_3$ point for both robots?

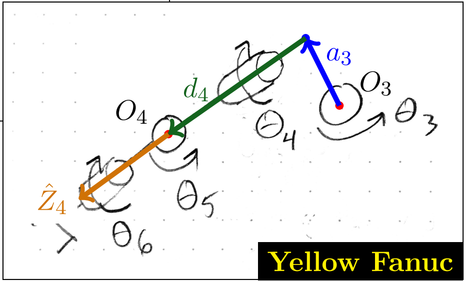
Comparing the White and Yellow Fanucs



- what do we call these distances?

Comparing the White and Yellow Fanucs

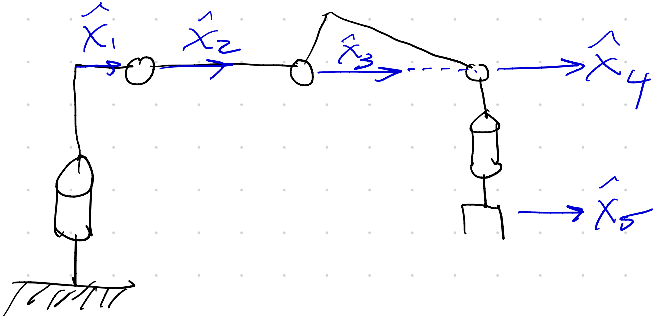




- what do we call these distances?

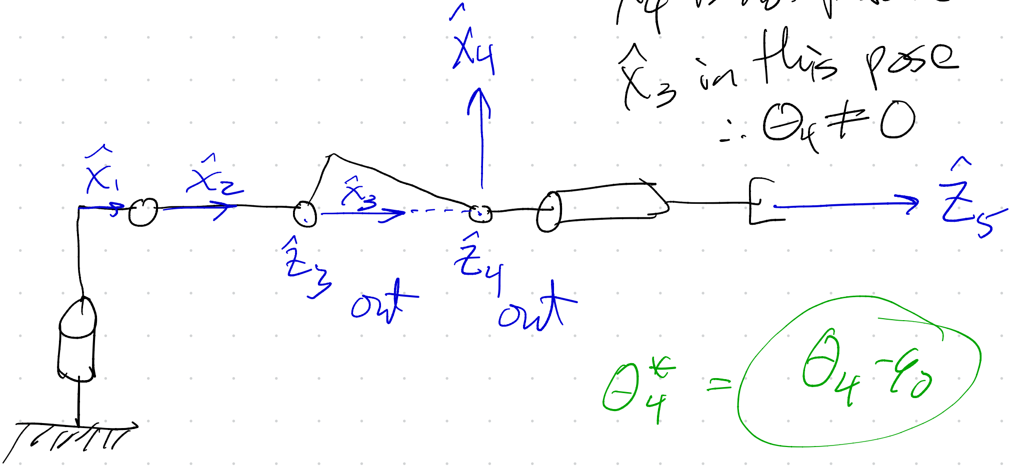
Draw the White Fanuc (5DoF) in the home position w/ joint offsets

all $\hat{X}_i$ are parallel



Can this also be a valid home position for the 5DOF Fanuc?

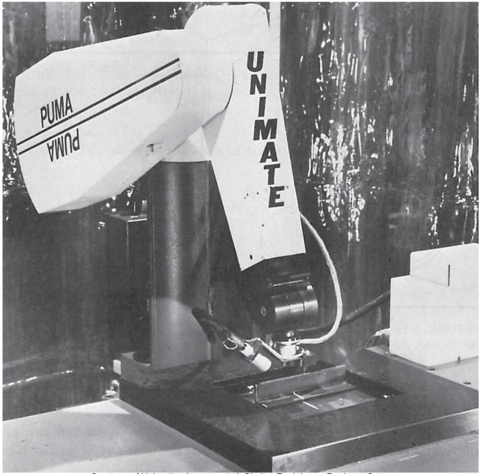
$\hat{X}_4$ is not parallel to $\hat{X}_3$ in this pose $\therefore \theta_4 \neq 0$



$$\theta_4^* = \theta_4 - \varphi_0$$

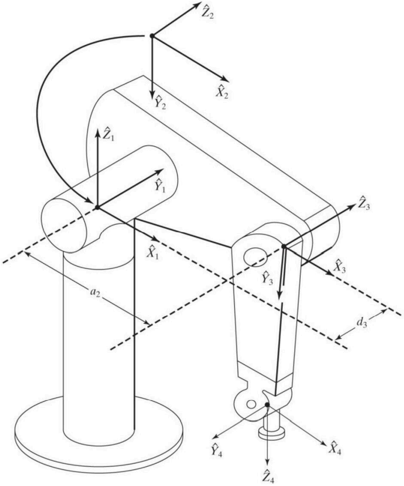
PUMA 560

- section 3.7, starting on page 83



Courtesy of Unimation Incorporated, Shelter Rock Lane, Danbury, Conn.

PUMA Main Schematic (Fig. 3.18)

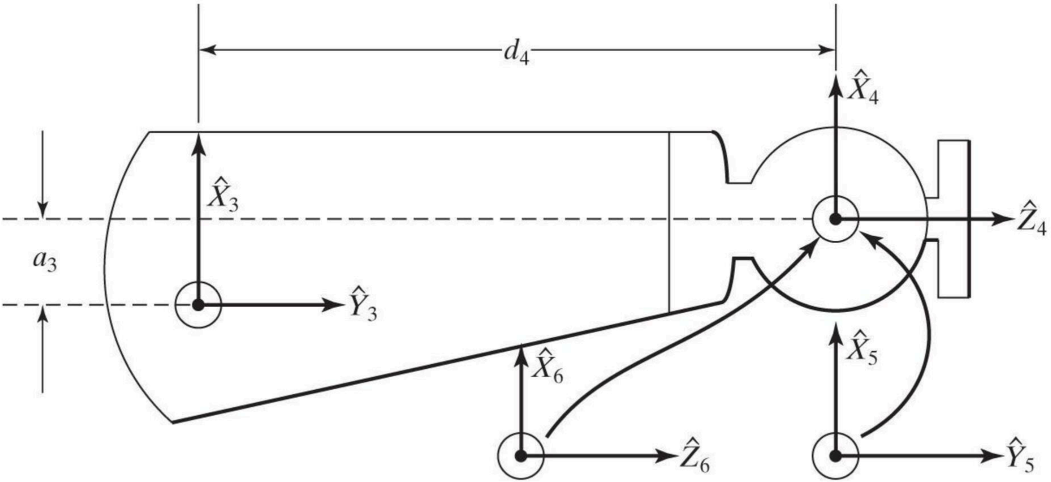


YouTube Animation

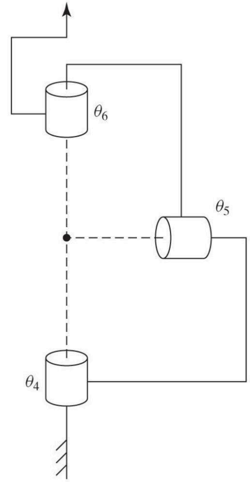
From Ohio University:

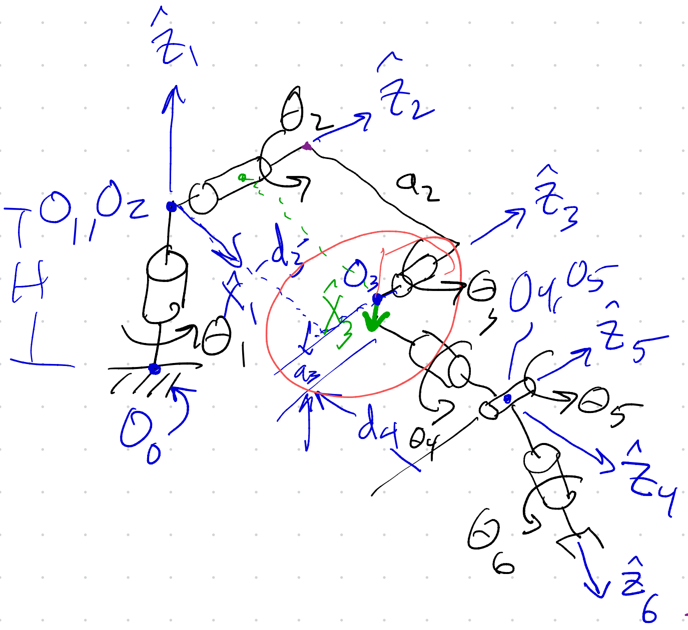
https://www.youtube.com/watch?v=ArzP7rh4_9Q

PUMA Forearm (Fig. 3.19)



PUMA Wrist (Fig. 3.20)





i	α	a	θ	d
1	0	0	θ_1	H
2	-90	0	θ_2	0
3	0	a_2	$\theta_3 + 90$	d_3
4	+90	a_3	θ_4	d_4
5	-90	0	θ_5	0
6	+90	0	θ_6	0 OR d_6

?